

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.2007.....

B.E. II (ME)

3

MD - I

**SECOND B.E. EXAMINATION - 2025
MECHANICAL ENGINEERING
(IV SEMESTER CBCS)
(MAIN/ATKT/EX)
4ME44A: MACHINE DESIGN - I**

Time - Three Hours

Maximum Marks - 70

- Note:-** (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.
(iv) Use of design data hand book is permitted

1. (a) What important considerations are required to be taken into account while designing Cast and Forged parts? 7

(b) On what factors do the selection of raw material depends? 7

2. (a) Explain the concept of standardization and interchangeability. 7

- (b) Explain designing through morphological analysis. 7
3. Design a Knuckle joint for a tie rod of a circular section to sustain a maximum pull of 70 kN. The ultimate tensile strength of the rod material is 420 N/mm^2 . The ultimate tensile and shearing strength of pin material are respectively 720 N/mm^2 and 390 N/mm^2 . A factor of safety of 5 is to be used. The permissible stresses in tension and compression are equal. 14
4. Design a flat belt and pulley to transmit 8 kW power. The motor speed is 1440 rpm and compressor speed is 670 rpm. 14
5. Design a bell-crank lever of I-section to carry a load of 1000 N. Use C.I. for the lever with permissible tensile stress 20 MPa and shear stress 60 MPa, Bearing pressure for pin material 20 MPa. Take length of load arm and effort arm are 250 mm and 400 mm respectively. Also design the fulcrum pin assuming suitable connection type. 14

6. A truck spring has 12 number of leaves, two of which are full length leaves. The spring supports are 1.05 m apart and the central band is 85 mm wide. The central load is to be 5.4 kN with a permissible stress of 280 MPa. Determine the thickness and width of the steel spring leaves. The ratio of the total depth to width ratio is 3. Also determine the deflection of the spring. 14

7. A cast iron flange coupling for a 5.5 cm diameter shaft is fastened to it by means of rectangular sunk key and is provided with 4 bolt holes at circle diameter of 14 cm for fastening it to other half of the coupling. Making suitable assumption and assuming that the strength of the key & bolts is 77% of that of the material of the shaft calculate the dimensions of the key, bolts and coupling. 14

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Kin. & Dyn. of Mac. (M)

SECOND B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

(MAIN/ATKT/EX)

**4ME - 43A: KINEMATICS & DYNAMICS OF
MACHINES (M)**

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.
(iv) Assume suitable data, if not given and mention it clearly.

1. (a) Explain about a double slider crank chain and its inversions. The distance between two parallel shafts is 18mm and they are connected by an Oldham's coupling. The driving shaft revolves at 160 rpm. What will be the maximum speed of sliding of the tongue of the intermediate piece along the groove? 7

- (b) Describe the geometry of crank and slotted link mechanism used in shaper machines. Given the crank length is 120 mm and the slotted link has an effective radius of 400 mm, determine the stroke length and quick return ratio, assuming crank rotates 180° for on stroke. 7

2. (a) Explain the working principle of a prony brake dynamometer with a neat sketch. A prony brake is used to measure the power output of a 4 kW engine. The effective radius of the brake drum is 250 mm, and the difference in spring balance reading is 40N. Find the speed of the engine in rpm. 7

- (b) Define a cam and follower mechanism. Classify different types of cams and followers with neat sketches. 7

3. (a) A cam rotating at 1000 rpm gives a uniformly accelerated and retarded motion to a roller follower. The lift is 40 mm in 120° of rotation.

- (i) Plot the displacement, velocity and acceleration curves.
- (ii) Calculate : Time of rise, maximum velocity, maximum acceleration. 14

4. (a) Describe the working of a proell governor. A porter governor running at 200 rpm changes its sleeve position by 20 mm for a speed change of $\pm 3\%$. Calculate the effort and power of the governor. Assume sleeve weight = 25 kg. 7
- (b) A punching press is desired to punch 720 hole/hr. Each hole diameter is 2mm and thickness of sheet is 3 mm. It requires 20 Joules of energy per mm^3 of sheared area. If each punching operation takes $1/4$ second and the speed of fly wheel is reduced from 100 rpm to 80 rpm during punching. Find the mass of flywheel required in order to have desired punching operation if radius of gyration of flywheel is 30 cm. 7
5. (a) Discuss the role of gyroscopic couple in determining the stability of road vehicles. A car engine weighing 200 kg has a rotating part with moment of inertia $1.2 \text{ kg} \cdot \text{m}^2$ and runs at 3600 rpm. The car takes a right turn with a turning radius of 20 m at 60 km/hr. Calculate the gyroscopic couple and its effect. 7

- (b) A ship is turning left with a speed of 30 km/hr and a turning radius of 100 m. A rotor of mass 500 kg rotates at 3000 rpm clockwise when viewed from the front. The rotor has radius of gyration of 0.5 m.
- Calculate the gyroscopic couple acting on the ship.
 - State the effect of this couple on the pitching or yawing motion of the ship. 7
6. (a) An engine develops a turning moment $T = 5000 + 1500 \sin\theta - 1200 \cos 2\theta$ Nm. The flywheel has a moment in inertia of $25 \text{ kg} \cdot \text{m}^2$ and runs at 300 rpm.
- Draw the turning moment diagram
 - Determine the fluctuation of energy
 - Calculate the maximum angular acceleration. 7
- (b) Describe the double block brake. Why is it used instead of a single block in some applications? Derive the torque equation and explain the condition for self locking. 7
7. Write short note on any of two of the following:
- Principle of virtual work
 - Effect of braking on four-wheeler vehicles
 - Watt governor 2X7=14

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MCM

SECOND B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(IV SEMESTER)

(MAIN/ATKT/ÉX)

4ME42A (ME): METAL CUTTING &

METROLOGY

Time - Three Hours

Maximum Marks - 70

- Note:-** (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Explain the mechanism of chip formation in metal cutting. Define the term "chip flow" and discuss its significance in machining operations. 7
- (b) Compare the chip formation mechanisms in orthogonal and oblique cutting. How does the cutting force system differ between orthogonal and oblique cutting? 7

2.

In an orthogonal cutting operation, the following data is recorded.

Cutting force (F_c) = 1500 N

Thrust force (F_t) = 600 N

Rake angle (α) = 10°

Uncut chip thickness (t_1) = 0.3 mm

Width of cut (b) = 4 mm

- (a) Draw Merchant's Circle Diagram and label all forces (F_s , F_n , F , N).
- (b) Derive the expression for shear force (F_s) and calculate its value.
- (c) Explain the physical significance of the coefficient of friction (μ) in this context and compute it.
- (d) Determine the shear angle (ϕ) if the chip thickness ratio (r) is 0.5.

14

3.

- (a) Discuss in details the Taylor's tool life equation and explain the significance of each term (V , T , n , C). If two tools have Taylor's exponents $n_1=0.2$ and $n_2=0.5$, which tool will last longer at the same cutting speed?

7

- (b) List the essential properties for an ideal cutting tool material. Explain each property briefly. Explain how cutting speed, feed, and depth of cut affect the performance of different tool materials.

7

4. (a) Define Metrology and explain its significance in manufacturing and quality control. Differentiate between dimensional metrology and geometrical metrology with suitable examples.

7

- (b) Differentiate among Mechanical, Electrical, Optical and Pneumatic Comparators in terms of operating principle, working parameters, advantages and limitations.

7

5. (a) How is an optical flat used to measure flatness? Explain with a diagram. Describe the use of autocollimators for parallelism measurement in large machine beds.

7

- (b) Explain the significance of thread fit (e.g., clearance, transition, interference) in screw thread applications. Why is accurate measurement of screw threads critical in mechanical assemblies?

7

6. (a) What is machine tools metrology, and why are alignment tests essential for machining accuracy? Define geometric accuracy and kinematic accuracy in machine tools. How do they differ? 7
- (b) Describe the major alignment tests performed on machine tools (e.g., lathe, milling machine, grinding machine). Explain how misalignment in machine tools affects workpiece quality. 7
7. (a) Define variability in quality control. Why is understanding? Differentiate between assignable causes and unassignable causes of variability. 7
- (b) Define control charts and explain their significance in statistical process control (SPC). Differentiate between control charts for variables and attributes with examples. 7
8. (a) What is an Operating Characteristic (OC) curve? How does it help in evaluating a sampling plan? Explain the terms Acceptable Quality Level (AQL) and Lot Tolerance Percent Defective (LTPD) in acceptance sampling. 7
- (b) What are the key benefits of implementing Total Quality Management in an organization? Explain the basic tools of quality used in TQM. 7